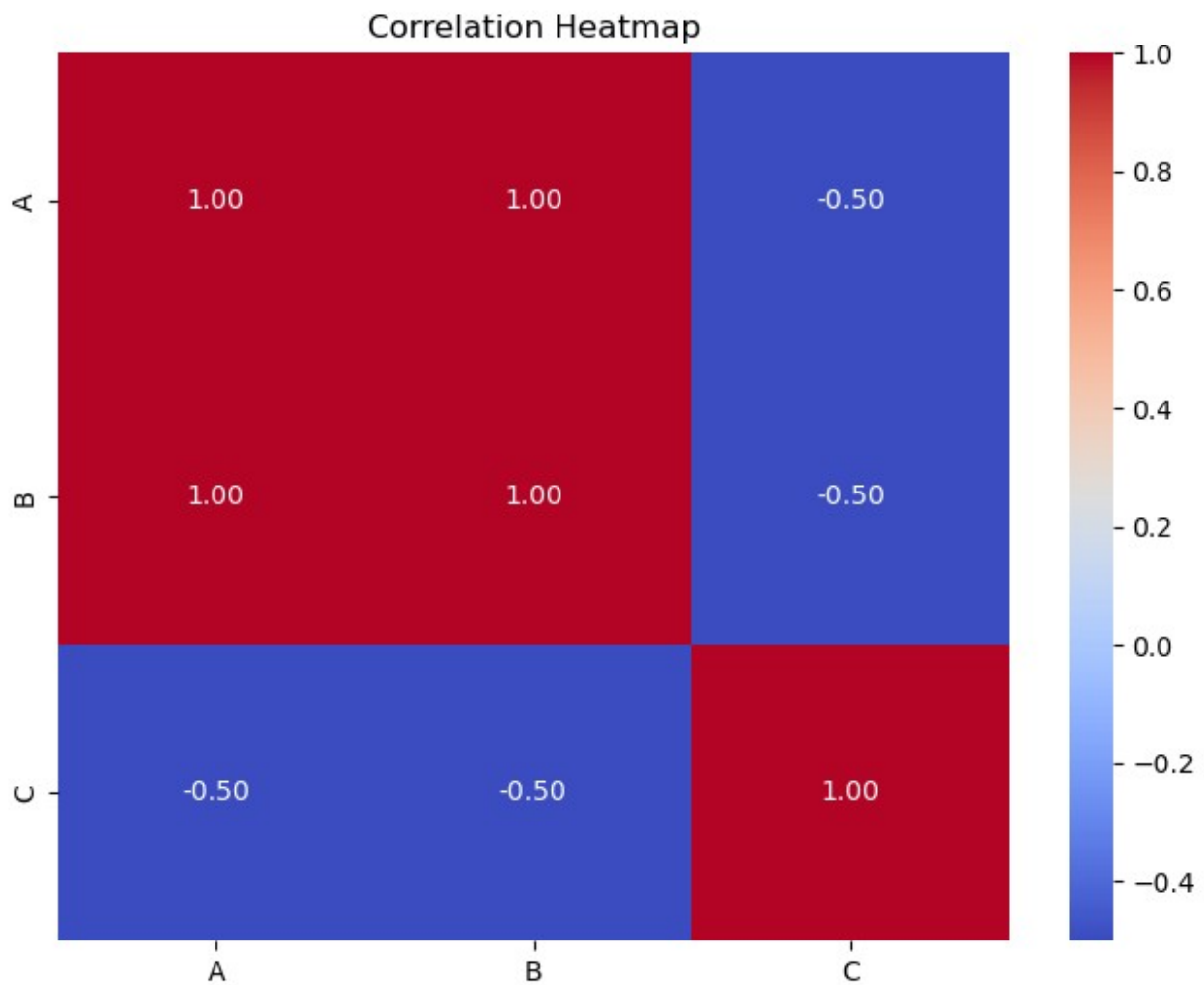


```

import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
df = pd.DataFrame({
    'A': [1,2,3,4,5],
    'B': [2,4,6,8,10],
    'C': [5,3,1,4,2]
})
plt.figure(figsize=(8,6))
corr = df.corr(numeric_only=True)
sns.heatmap(
    corr,
    annot=True,
    cmap='coolwarm',
    fmt='.2f'
)
plt.title("Correlation Heatmap")
plt.show()

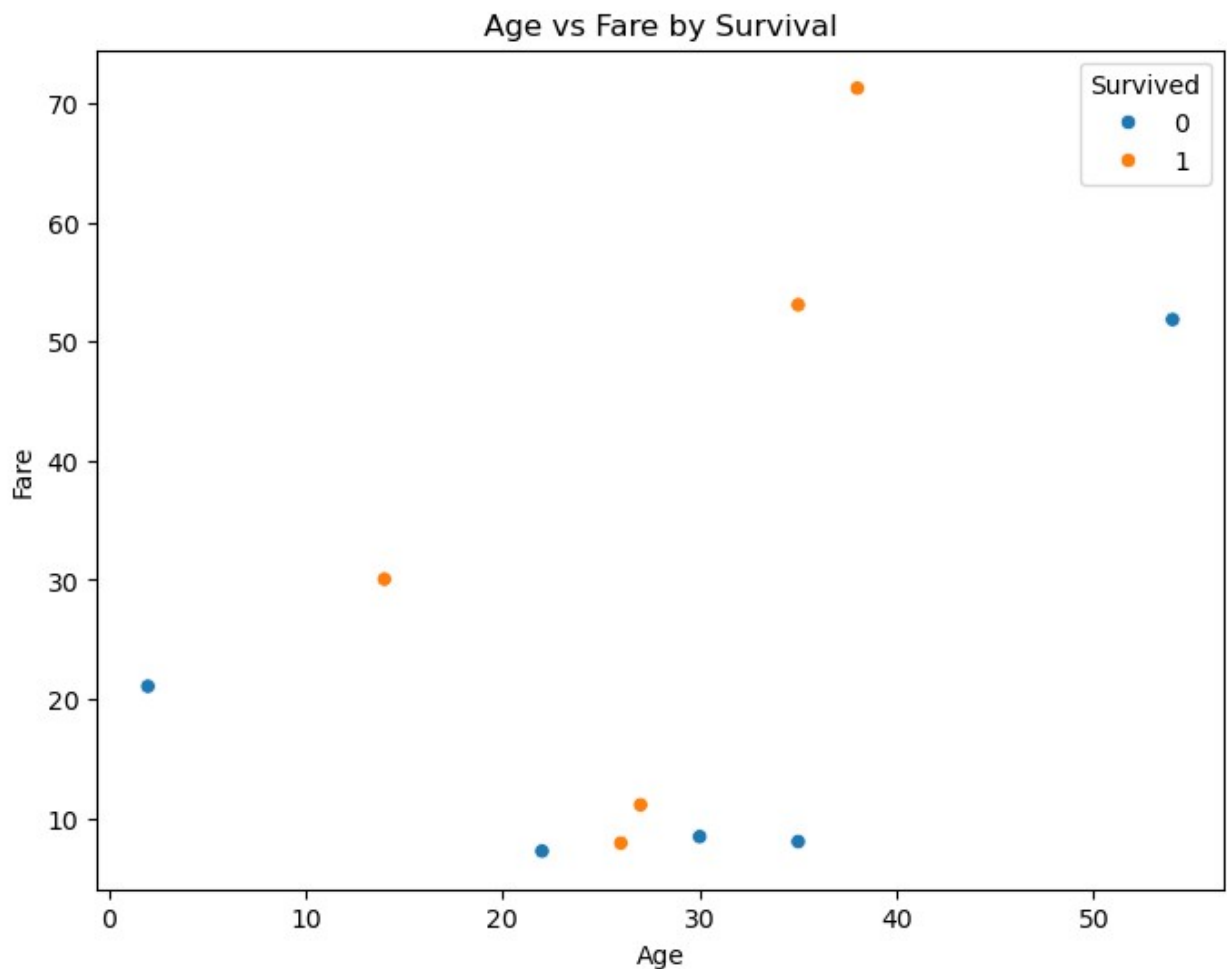
```



```

import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
df = pd.DataFrame({'Age': [22, 38, 26, 35, 35, 30, 54, 2, 27, 14], 'Fare': [7.25, 71.28, 7.92, 53.1, 8.05, 8.46, 51.86, 21.08, 11.13, 30.07], 'Survived': [0, 1, 1, 1, 0, 0, 0, 0, 1, 1]})
plt.figure(figsize=(8,6))
sns.scatterplot(x='Age', y='Fare', hue='Survived', data=df)
plt.title("Age vs Fare by Survival")
plt.xlabel("Age"); plt.ylabel("Fare")
plt.show()

```



```

import numpy as np
data = [5, 10, 15, 20, 25, 30, 35, 40]
Q1 = np.percentile(data, 25)
Q3 = np.percentile(data, 75)
IQR = Q3 - Q1
print("Q1:", Q1)

```

```
print("Q3:", Q3)
print("IQR:", IQR)
```

```
Q1: 13.75
Q3: 31.25
IQR: 17.5
```

```
import numpy as np
data = [10, 12, 14, 15, 16, 18, 100]
Q1 = np.percentile(data, 25)
Q3 = np.percentile(data, 75)
IQR = Q3 - Q1
lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR
print("Lower bound:", lower)
print("Upper bound:", upper)
outliers = [x for x in data if x < lower or x > upper]
print("Outliers:", outliers)
```

```
Lower bound: 7.0
Upper bound: 23.0
Outliers: [100]
```

```
sample_space = {1, 2, 3, 4, 5, 6}
event_A = {2, 4, 6} # Even numbers
print("Sample Space (S):", sample_space)
print("Event A:", event_A)
P_A = len(event_A) / len(sample_space)
print("\nProbability of Event A")
print("P(A) =", P_A)
```

```
Sample Space (S): {1, 2, 3, 4, 5, 6}
Event A: {2, 4, 6}
```

```
Probability of Event A
P(A) = 0.5
```

```
event_B = {4, 5, 6}
intersection = event_A.intersection(event_B)
P_A_given_B = len(intersection) / len(event_B)
print("\nEvent B:", event_B)
print("A ∩ B:", intersection)
print("P(A|B) =", P_A_given_B)
```

```
Event B: {4, 5, 6}
A ∩ B: {4, 6}
P(A|B) = 0.6666666666666666
```

```
P_Disease = 0.01
P_Positive_given_Disease = 0.99
```

```

P_Positive = 0.05
P_Disease_given_Positive = (
    P_Positive_given_Disease * P_Disease
) / P_Positive
print("P(Disease|Positive) =", P_Disease_given_Positive)
print(
    "P(Disease | Positive) =",
    round(P_Disease_given_Positive, 4)
)
print(
    "Percentage =",
    round(P_Disease_given_Positive * 100, 2),
    "%"
)

```

```

P(Disease|Positive) = 0.198
P(Disease | Positive) = 0.198
Percentage = 19.8 %

```

```

import pandas as pd
df = pd.DataFrame({
    'Hours': [1,2,3,4,5],
    'Marks': [20,40,50,70,90]
})
corr = df['Hours'].corr(df['Marks'])
print("Pearson Correlation:", corr)

Pearson Correlation: 0.9948497511671097

```

```

import pandas as pd
df = pd.DataFrame({
    'Rank_Study': [1, 2, 3, 4, 5],
    'Rank_Marks': [2, 1, 3, 5, 4]
})
corr = df['Rank_Study'].corr(df['Rank_Marks'], method='spearman')
print("Spearman Correlation:", corr)

Spearman Correlation: 0.7999999999999999

```

```

import pandas as pd
df = pd.DataFrame({
    'Hours':
    [1,2,3,4,5],
    'Marks':
    [20,40,50,70,90],
    'Sleep': [8,7,6,5,4]
})
print(df.corr())
sns.heatmap(df.corr(),
annot=True,

```

```
cmap='plasma')  
plt.show()
```

	Hours	Marks	Sleep
Hours	1.00000	0.99485	-1.00000
Marks	0.99485	1.00000	-0.99485
Sleep	-1.00000	-0.99485	1.00000

